

Kaichen Xu

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EDUCATION

Emory University

Sep 2025 – Present

Major: Master of Science in Computer Science **GPA:** 3.90/4.00

Zhongnan University of Economics and Law (ZUEL)

Sep 2020 – Jun 2024

Major: Bachelor of Science in Statistics **GPA:** 3.85/4.00 (92.21/100)

PUBLICATIONS

*: equal contribution

- [1] **Kaichen Xu***, Yuxuan Shen*, et al. Building a causality-aware single-cell RNA-seq foundation model via context-specific causal regulation modeling. *Nature Biomedical Engineering*, 2026.
- [2] **Kaichen Xu**, Yuxuan Shen, et al. Bringing cell-specific regulatory logic into foundation models. *Nature Biomedical Engineering*, 2026.
- [3] **Kaichen Xu**, Yan Lu, et al. Detecting and dissecting anomalous anatomic regions in spatial transcriptomics with STANDS. *Nature Communications*, 2024.
- [4] **Kaichen Xu**, Mianpeng Liu, et al. Concord: building consensus representations for single cells with collaborative random projection. In *Proceedings of the SIGKDD Conference on Knowledge Discovery and Data Mining (KDD)*, 2026.
- [5] **Kaichen Xu**, Yihang Du, et al. Causality-induced positional encoding for transformer-based representation learning of non-sequential features. In *Proceedings of the International Conference on Neural Information Processing Systems (NeurIPS)*, 2025.
- [6] **Kaichen Xu***, Qilong Wu*, et al. Multimodal anomalous tissue region detection enhanced with spatial transcriptomics. In *Proceedings of the AAAI Conference on Artificial Intelligence (AAAI)*, 2025.
- [7] **Kaichen Xu***, Yueyang Ding*, et al. Domain adaptive and fine-grained anomaly detection for single-cell sequencing data and beyond. In *Proceedings of the International Joint Conference on Artificial Intelligence (IJCAI)*, 2024.
- [8] **Kaichen Xu***, Kainan Liu*, et al. Detecting and subtyping anomalous single cells with M2ASDA. In *ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM BCB)*, 2024.
- [9] Tao Deng, Mengqian Huang, **Kaichen Xu**, et al. LEGEND: an integrative algorithm for identifying co-expressed and cofunctional genes in multimodal transcriptomic sequencing data. *Genomics Proteomics & Bioinformatics*, 2025.

RESEARCH EXPERIENCES

Building a causality-aware single-cell RNA-seq foundation model via context-specific causal regulation modeling

- Developed scCAFM, a foundation model pretrained on 49.7 million cells that integrates cell-specific gene regulatory network (GRN) inference with causality-aware representation learning.
- Enabled scalable GRN inference across thousands of genes and millions of cells, capturing regulatory heterogeneity and developmental dynamics.
- Designed autoregressive learning guided by inferred regulatory networks to generate gene and cell representations that preserve context-specific regulatory structure.
- Demonstrated superior performance across gene perturbation prediction, drug response prediction, and cell type annotation, and enabled prediction of perturbation-induced cell fate transitions.
- Published in Nature Biomedical Engineering (accepted).

Causality-induced positional encoding for transformer-based representation learning of non-sequential features

- Investigated the theoretical relationship between sequential ordering and causal dependency structures.
- Proposed CAPE (Causality-Aware Positional Encoding), a framework integrating causal discovery with transformer architectures to generalize positional encoding beyond sequential inputs.
- Enabled attention mechanisms to capture causal strength and structural dependencies among unordered features, improving representation learning in tabular and structured datasets.

- Conducted theoretical analysis and large-scale empirical evaluations demonstrating significant improvements over standard positional encoding strategies.
- Published at [NeurIPS 2025](#).

Detecting and dissecting anomalous anatomic regions in spatial transcriptomics with STANDS

- Studied disease heterogeneity and microenvironmental variation in spatial transcriptomics data.
- Identified key challenges in population-level versus patient-specific pathological patterns affecting disease characterization.
- Proposed STANDS, a multimodal deep learning framework based on Generative Adversarial Networks integrating spatial gene expression profiles with histopathology (H&E) imaging.
- Achieved state-of-the-art performance in detecting pathological tissue regions across multiple spatial transcriptomics datasets.
- Published in [Nature Communications](#) (undergraduate first author).

WORK EXPERIENCES

Emory University **Atlanta, GA**

Research Assistant (Advisor: [Prof. Victor Corces](#)) Mar 2026 – Present

- Conduct research on alternative splicing and RNA regulation using multimodal single-cell genomics data.
- Investigate how sequence context and chromatin accessibility jointly regulate RNA processing, providing a predictive framework for alternative splicing across cell types.
- Led a research project as co-first author, preparing the manuscript for submission to Nature Genetics.

The Chinese University of Hong Kong (Shenzhen) **Shenzhen, China**

Research Assistant (Advisor: [Prof. Jin Liu](#)) Jun 2025 – Dec 2025

- Conducted research on gene regulatory network (GRN) inference from single-cell transcriptomic data, aiming to uncover regulatory interactions among genes in complex biological systems.
- Developed scalable and interpretable single cell foundation models for inferring gene-gene regulatory relationships.
- Led a research project as first author; the resulting manuscript is currently under review at Nature Biomedical Engineering.

Zhongnan University of Economics and Law **Wuhan, China**

Research Assistant (Advisor: [Prof. Xiaobo Sun](#)) Mar 2023 – Dec 2025

- Conducted research on anomaly detection, multimodal biological data integration, and deep learning theory.
- Mentored junior undergraduate researchers in machine learning and computational biology, providing guidance on research design, experimentation, and paper writing; mentees subsequently achieved publications in major AI conferences.
- Research outcomes resulted in first-author publications in Nature Communications, IJCAI, AAAI, and NeurIPS.

LEADERSHIP & MENTORSHIP

Organizer and Speaker — Machine Learning Academic Seminar Group

- Founded a weekly online seminar on machine learning research, engaging over 30 participants from multiple universities.
- Delivered monthly presentations on recent ML papers, facilitating discussions on emerging research directions.
- Mentored participants in research methodology and project development; members later produced 10+ conference papers in AI and computational biology venues.

Server Administrator and Research Mentor — Sunlab

- Managed the lab's servers and computing infrastructure, ensuring reliable access for large-scale machine learning experiments.
- Conducted 30+ interviews for prospective lab members and contributed to research group recruitment.
- Mentored junior researchers in machine learning and experimental design; one mentee published a paper at IJCAI 2025 as an undergraduate.

INVITED TALKS & PRESENTATIONS

Oral Presentation at ACM SIGKDD Conference on Knowledge Discovery and Data Mining (BioKDD workshop)	Aug 2026
Oral Academic Seminar for introducing our work STANDS, Peking University	Dec 2024
Oral Presentation at ACM Conference on Bioinformatics, Computational Biology, and Health Informatics (ACM BCB)	Nov 2024
Oral Presentation at International Joint Conference on Artificial Intelligence (IJCAI)	Aug 2024